

3.1 (a) $F(x, y, z) = \Sigma(0, 2, 3, 7)$

$\begin{array}{c} yz \\ \diagdown \\ x \end{array}$		y			
		00	01	11	10
x	0	m_0 1	m_1	m_3 1	m_2 1
	1	m_4	m_5	m_7 1	m_6

$$F = x'z' + yz$$

(b) $F(x, y, z) = \Sigma(2, 3, 5, 6, 7)$

$\begin{array}{c} yz \\ \diagdown \\ x \end{array}$		y			
		00	01	11	10
x	0	m_0	m_1	m_3 1	m_2 1
	1	m_4	m_5 1	m_7 1	m_6 1

$$F = y + xz$$

(c) $F(x, y, z) = \Sigma(0, 1, 2, 4, 6)$

$\begin{array}{c} yz \\ \diagdown \\ x \end{array}$		y			
		00	01	11	10
x	0	m_0 1	m_1 1	m_3	m_2 1
	1	m_4 1	m_5	m_7	m_6 1

$$F = z' + x'y'$$

(d) $F(x, y, z) = \Sigma(1, 3, 5, 7)$

$\begin{array}{c} yz \\ \diagdown \\ x \end{array}$		y			
		00	01	11	10
x	0	m_0	m_1 1	m_3 1	m_2
	1	m_4	m_5 1	m_7 1	m_6

$$F = z$$

3.2 (a) $F(x, y, z) = \Sigma(2, 3, 4, 5, 7)$

		y			
		00	01	11	10
x \ yz	0	m_0	m_1	m_3 1	m_2 1
	1	m_4 1	m_5 1	m_7 1	m_6

$$F = x'y + xy' + xz$$

(b) $F(x, y, z) = \Sigma(0, 1, 4, 5, 6, 7)$

		y			
		00	01	11	10
x \ yz	0	m_0 1	m_1 1	m_3	m_2
	1	m_4 1	m_5 1	m_7 1	m_6 1

$$F = x + y'$$

(c) $F(x, y, z) = \Sigma(0, 1, 2, 4, 5, 6)$

		y			
		00	01	11	10
x \ yz	0	m_0 1	m_1 1	m_3	m_2 1
	1	m_4 1	m_5 1	m_7	m_6 1

$$F = y' + z'$$

(d) $F(x, y, z) = \Sigma(1, 2, 3, 5, 6, 7)$

		y			
		00	01	11	10
x \ yz	0	m_0	m_1 1	m_3 1	m_2 1
	1	m_4	m_5 1	m_7 1	m_6 1

$$F = y + z$$

(e) $F(x, y, z) = \Sigma(1, 3, 5, 7)$

		y			
		00	01	11	10
x \ yz	0	m_0	m_1 1	m_3 1	m_2
	1	m_4	m_5 1	m_7 1	m_6

$$F = z$$

(f) $F(x, y, z) = \Sigma(0, 1, 2, 3, 5, 7)$

		y			
		00	01	11	10
x \ yz	0	m_0 1	m_1 1	m_3 1	m_2 1
	1	m_4	m_5 1	m_7 1	m_6

$$F = x' + z$$

3.3 (a) $F(x, y, z) = x'y'z + x'yz' + xyz' + xy$

		y			
		00	01	11	10
x	yz				
		m_0	m_1	m_3	m_2
0			1	1	
1		m_4	m_5	1	m_6
				1	1

$$F = x'z + xy$$

(b) $F(x, y, z) = xyz' + y'z' + xz$

		y			
		00	01	11	10
x	yz				
		m_0	m_1	m_3	m_2
0		1			
1		m_4	m_5	1	m_6
		1	1	1	1

$$F = x + y'z'$$

(c) $F(x, y, z) = x'yz + xy' + yz'$

		y			
		00	01	11	10
x	yz				
		m_0	m_1	m_3	m_2
0				1	1
1		m_4	m_5	m_7	m_6
		1	1		1

$$F = xy' + x'yz + yz'$$

(d) $F(x, y, z) = xz' + x'z + x'y$

		y			
		00	01	11	10
x	yz				
		m_0	m_1	m_3	m_2
0			1	1	1
1		m_4	m_5	m_7	m_6
		1	1		1

$$F = xz' + xy' + yz'$$

3.4 (a) $F(w, x, y, z) = \Sigma(0, 1, 4, 5, 8, 12)$

			y			
			00	01	11	10
w	xz	yz				
			m_0	m_1	m_3	m_2
w	00		1	1		
	01		m_4	m_5	m_7	m_6
	11		m_{12}	m_{13}	m_{15}	m_{14}
	10		m_8	m_9	m_{11}	m_{10}

$$F = y'z' + w'y'$$

(b) $F(w, x, y, z) = \Sigma(4, 6, 9, 11, 12, 14)$

			y			
			00	01	11	10
w	xz	yz				
			m_0	m_1	m_3	m_2
w	00					
	01		m_4	m_5	m_7	m_6
	11		m_{12}	m_{13}	m_{15}	m_{14}
	10		m_8	m_9	m_{11}	m_{10}

$$F = xz' + wx'z$$

(c) $F(w, x, y, z) = \Sigma(1, 3, 5, 6, 7, 9, 11)$ (d) $F(w, x, y, z) = \Sigma(2, 6, 7, 10, 14, 15)$

			y			
			00	01	11	10
w	xz	yz				
			m_0	m_1	m_3	m_2
w	00			1	1	
	01		m_4	m_5	m_7	m_6
	11		m_{12}	m_{13}	m_{15}	m_{14}
	10		m_8	m_9	m_{11}	m_{10}

$$F = w'z + x'z + w'xy$$

			y			
			00	01	11	10
w	xz	yz				
			m_0	m_1	m_3	m_2
w	00					1
	01		m_4	m_5	m_7	m_6
	11		m_{12}	m_{13}	m_{15}	m_{14}
	10		m_8	m_9	m_{11}	m_{10}

$$F = yz' + xy$$

(e) $F(w, x, y, z) = \Sigma(9, 10, 11, 12, 13, 15)$ (f) $F(w, x, y, z) = \Sigma(0, 2, 4, 6, 9, 11, 13, 15)$

		y			
		00	01	11	10
w	yz \ wx				
	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F = wx'y' + wz + wx'y$$

		y			
		00	01	11	10
w	yz \ wx				
	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F = w'z' + wz$$

(g) $F(w, x, y, z) = \Sigma(0, 1, 2, 4, 6, 8, 9)$ (h) $F(w, x, y, z) = \Sigma(0, 3, 6, 7, 8, 11, 14, 15)$

		y			
		00	01	11	10
w	yz \ wx				
	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F = w'z' + x'y'$$

		y			
		00	01	11	10
w	yz \ wx				
	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F = x'y'z' + yz + xy$$

3.5 (a) $F(w, x, y, z) = \Sigma(0, 2, 3, 4, 6, 8, 9, 12)$ (b) $F(w, x, y, z) = \Sigma(0, 1, 2, 3, 5, 8, 13)$

		y			
		00	01	11	10
w	yz \ wx				
	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F = y'z' + wx'y' + w'x'y + w'yz'$$

		y			
		00	01	11	10
w	yz \ wx				
	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F = x'y'z' + w'x'z + xy'z$$

(c) $F(w, x, y, z) = \Sigma(2, 3, 5, 7, 11, 13)$ (d) $F(w, x, y, z) = \Sigma(1, 2, 3, 6, 7, 8, 9, 12, 13)$

		y			
		00	01	11	10
w	yz \ wx				
	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F = w'x'y + w'xz + xy'z + x'yz$$

		y			
		00	01	11	10
w	yz \ wx				
	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F = w'x'z + wy' + w'y$$

3.6 (a) $ABC'D' + AB'C + B'C'D + AB'CD + B'C'D$ (b) $w'x'y' + w'x'y + x'y'z + xyz + y'z'$

		C			
		00	01	11	10
A	00	m ₀ 1	m ₁ 1	m ₂	m ₃
	01	m ₄	m ₅	m ₆	m ₇
	11	m ₁₂ 1	m ₁₃	m ₁₅	m ₁₄
	10	m ₈ 1	m ₉	m ₁₁ 1	m ₁₀ 1

$$F = AC'D' + B'C' + AB'$$

		y			
		00	01	11	10
w	00	m ₀ 1	m ₁ 1	m ₂ 1	m ₃
	01	m ₄ 1	m ₅	m ₇	m ₆
	11	m ₁₂ 1	m ₁₃	m ₁₅	m ₁₄
	10	m ₈ 1	m ₉	m ₁₁ 1	m ₁₀

$$F = y'z' + x'y' + w'x'z$$

(c) $A'B'CD + A'BC + C'D + ABCD + AB'C$

		C			
		00	01	11	10
A	00	m ₀	m ₁ 1	m ₂ 1	m ₃
	01	m ₄	m ₅ 1	m ₆ 1	m ₇ 1
	11	m ₁₂	m ₁₃ 1	m ₁₅ 1	m ₁₄
	10	m ₈	m ₉ 1	m ₁₁ 1	m ₁₀ 1

$$F = D + A'BC + AB'C$$

(d) $A'B'C' + A'BD + A'BC'D' + BC'D + ABCD$

		C			
		00	01	11	10
A	00	m ₀ 1	m ₁ 1	m ₂	m ₃
	01	m ₄ 1	m ₅ 1	m ₇ 1	m ₆
	11	m ₁₂	m ₁₃ 1	m ₁₅ 1	m ₁₄
	10	m ₈	m ₉	m ₁₁	m ₁₀

$$F = A'C' + BD$$

3.7 (a) $w'x'y' + xy' + w'x + wxy$

		y			
		00	01	11	10
w	00	m ₀	m ₁ 1	m ₂ 1	m ₃
	01	m ₄	m ₅ 1	m ₇ 1	m ₆ 1
	11	m ₁₂ 1	m ₁₃ 1	m ₁₅ 1	m ₁₄ 1
	10	m ₈	m ₉	m ₁₁	m ₁₀

$$F = x + w'z$$

(b) $A'BD' + BDC + ABC' + BD + ABC$

		C			
		00	01	11	10
A	00	m ₀	m ₁	m ₂	m ₃
	01	m ₄ 1	m ₅ 1	m ₇ 1	m ₆ 1
	11	m ₁₂ 1	m ₁₃ 1	m ₁₅ 1	m ₁₄
	10	m ₈	m ₉	m ₁₁	m ₁₀

$$F = BC' + BD + A'BC$$

(c) $AB'C' + AC'D' + ABD + BC'D + A'BC'D'$

		C			
		00	01	11	10
A	00	m ₀	m ₁	m ₂	m ₃
	01	m ₄ 1	m ₅ 1	m ₇	m ₆
	11	m ₁₂ 1	m ₁₃ 1	m ₁₅ 1	m ₁₄
	10	m ₈ 1	m ₉ 1	m ₁₁	m ₁₀

$$F = BC' + AC' + ABD$$

(d) $wxyz + wx' + wx'y + wxy + w'yz'$

		y			
		00	01	11	10
w	00	m ₀	m ₁	m ₂	m ₃ 1
	01	m ₄	m ₅	m ₇	m ₆ 1
	11	m ₁₂	m ₁₃	m ₁₅ 1	m ₁₄ 1
	10	m ₈ 1	m ₉ 1	m ₁₁ 1	m ₁₀ 1

$$F = wx' + wy + y'z$$

3.8 (a) $w'x + xz + w\gamma z$

		y			
		00	01	11	10
w	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F(w, x, y, z) = \Sigma(4, 5, 6, 7, 11, 13, 15)$$

$$= w'xy'z' + w'xy'z + w'xyz' + wx'y z + wx\gamma y'z + wx\gamma yz$$

(b) $ABD' + BC'D + CD$

		C			
		00	01	11	10
A	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F(A, B, C, D) = \Sigma(3, 5, 7, 11, 12, 13, 14, 15)$$

$$= A'B'CD + A'BC'D + A'BCD + AB'CD + ABC'D' + ABC'D + ABCD' + ABCD$$

(c) $x'y'z' + xy + y'z$

		y			
		00	01	11	10
x	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

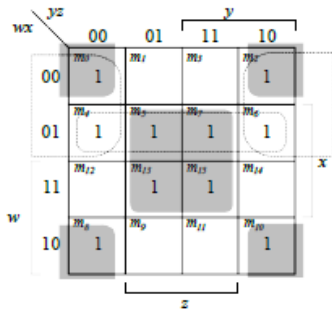
$$F(x, y, z) = \Sigma(0, 1, 5, 6, 7) = x'y'z' + x'y'z + xy'z + xyz' + xyz$$

(d) $AB + B'C'D' + BCD + A'C'D'$

		C			
		00	01	11	10
A	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$$F(A, B, C, D) = \Sigma(0, 4, 7, 8, 12, 13, 14, 15)$$

$$= A'B'C'D' + A'BC'D' + A'BCD + AB'C'D' + ABC'D' + ABC'D + ABCD' + ABCD$$



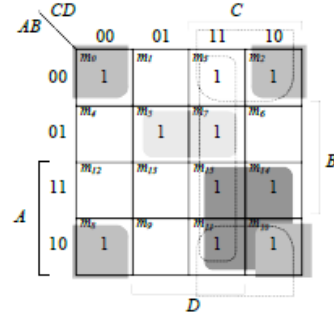
(a)

Essential: $xz, x'z'$

Non-essential: $w'x, w'z'$

$$F = xz + x'z' + (w'x \text{ or } w'z')$$

OR $B'C$)

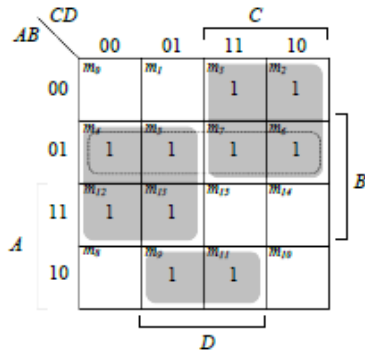


(b)

Essential: $B'D', AC, A'BD$

Non-essential: $CD, B'C$

$$F = B'D' + AC + A'BD + (CD$$

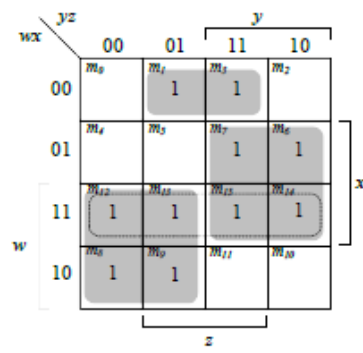


(c)

Essential: $BC', AC, A'B'D$

Non-Essential: $A'B$

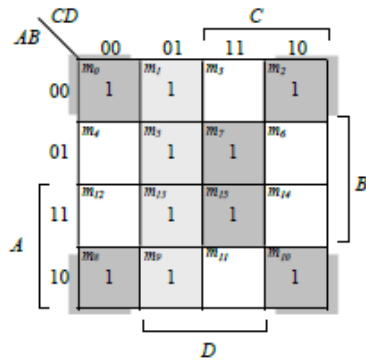
$$F = BC' + AC + A'B'D$$



(d)

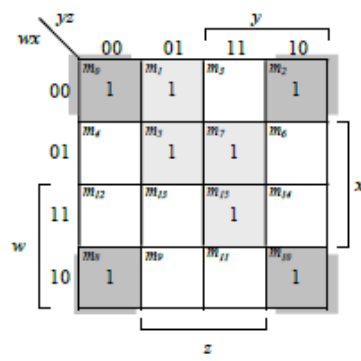
Essential: $wy', xy, w'x'z$

$$F = wy' + xy + w'x'z$$



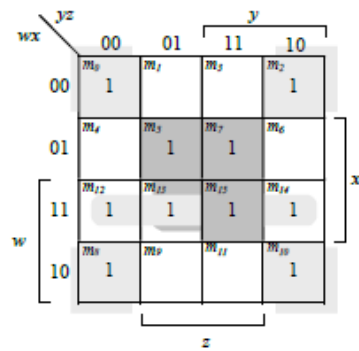
Essential: $BD, B'C', C'D$

$$F = BD + B'C' + C'D$$



Essential: $x'z', w'y'z, xyz$

$$F = x'z' + w'y'z + xyz$$

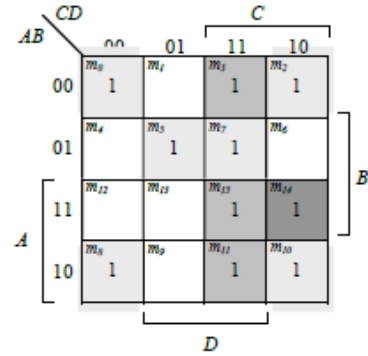


$$F = \Sigma(0, 2, 5, 7, 8, 10, 12, 13, 14, 15)$$

Essential: $xz, wx, x'z'$
 $A'BD$

$$F = xz + wx + x'z'$$

(a)

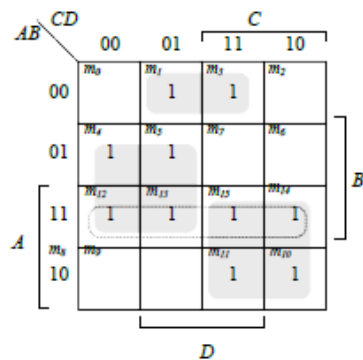


$$F = \Sigma(0, 2, 3, 5, 7, 8, 10, 11, 12, 13, 14, 15)$$

Essential: $AC, B'D', CD, A'BD$

$$F = AC + B'D' + CD + A'BD$$

(b)



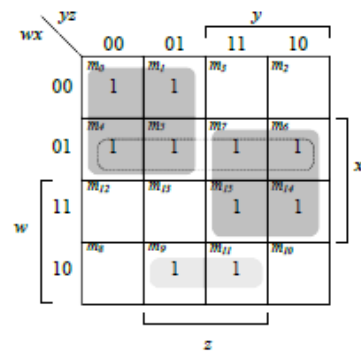
$$F = \Sigma(1, 3, 4, 5, 10, 11, 12, 13, 14, 15)$$

Essential: $AC, BC', A'B'D$

Non-essential: $AB, A \neq B \neq D, B \neq CD, A \neq C \neq D$
 $w'wz, w'x'z$

$$F = AC + BC' + A'B'D$$

(c)



$$F = \Sigma(0, 1, 4, 5, 6, 7, 9, 11, 12, 13, 14, 15)$$

Essential: $w'y', xy, wx'z$

Non-essential: $wx, x'y'z,$

$$F = w'y' + xy + wx'z$$

(d)

AB \ CD		C			
		00	01	11	10
A	00	m_0 1	m_1 1	m_3 1	m_2
	01	m_4 	m_5 	m_7 1	m_6
	11	m_{12} 	m_{13} 1	m_{15} 1	m_{14}
	10	m_8 1	m_9 1	m_{11} 	m_{10} 1

$$F(A, B, C, D) = \Sigma(0, 1, 3, 7, 8, 9, 10, 13, 15)$$

10, 15)

Essential: $B'C', AB'D'$

Non-essential: $ABD, A'CD, BCD$

$$F = B'C' + AB'D' + A'CD + ABD$$

(e)

wx \ yz		y			
		00	01	11	10
w	00	m_0 1	m_1 1	m_3 	m_2 1
	01	m_4 1	m_5 1	m_7 1	m_6 1
	11	m_{12} 	m_{13} 	m_{15} 1	m_{14}
	10	m_8 	m_9 	m_{11} 	m_{10} 1

$$F = \Sigma(0, 1, 2, 4, 5, 6, 7,$$

Essential: $w'y', w'z', xyz, x'yz'$

Non-Essential: $w'x$

$$F = w'y' + w'z' + xyz + x'yz'$$

(f)

3.11 K-map for $F(w, x, y, z) = \Sigma(1, 2, 4, 5, 9, 10, 13, 14)$

wx \ yz		y			
		00	01	11	10
w	00	m_0 	m_1 1	m_3 	m_2 1
	01	m_4 1	m_5 1	m_7 	m_6
	11	m_{12} 	m_{13} 1	m_{15} 	m_{14} 1
	10	m_8 	m_9 1	m_{11} 	m_{10} 1

$$F = y'z + w'xy' + w'yz' + x'yz'$$

K-map for $F'(w, x, y, z) = \Sigma(0, 3, 6, 7, 8, 11, 12, 15)$

wx \ yz		y			
		00	01	11	10
w	00	m_0 1	m_1 	m_3 1	m_2
	01	m_4 	m_5 	m_7 1	m_6 1
	11	m_{12} 1	m_{13} 	m_{15} 1	m_{14}
	10	m_8 1	m_9 	m_{11} 1	m_{10}

$$F = yz + w'xy + x'y'z' + wy'z'$$

$$= (y' + z')(w + x' + y')(x + y + z)(w' + y + z)$$

3.12 (a) $F(A, B, C, D) = \Pi(0, 1, 6, 7, 8, 10, 12, 14)$

$F'(A, B, C, D) = \Sigma(0, 1, 6, 7, 8, 10, 12, 14)$

$F(A, B, C, D) = \Sigma(2, 3, 4, 5, 9, 11, 13, 15)$

		y			
		00	01	11	10
w	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$F = w'x'y + w'xy' + wz$

(b) $F(A, B, C, D) = \Pi(0, 2, 5, 8, 10, 15)$

$F'(A, B, C, D) = \Sigma(0, 2, 5, 8, 10, 15)$

$F(A, B, C, D) = \Sigma(1, 3, 4, 6, 7, 9, 11, 12, 13, 14)$

		y			
		00	01	11	10
w	00	m_0	m_1	m_3	m_2
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	m_8	m_9	m_{11}	m_{10}

$F = xz' + x'z + wxy' + w'xy$

3.13 (a) $x'y'z + yz' + xy$

		y			
		00	01	11	10
x	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

$F = x'y' + xz$

		y			
		00	01	11	10
x	0	m_0	m_1	m_3	m_2
	1	m_4	m_5	m_7	m_6

$F' = x'z + xy'$

$F = (x + z')(x' + y)$

(b) $A'B + A'B'C + CD$

		C			
		CD		11	10
A	AB	00	01	11	10
		m_0	m_1	m_3	m_2
	00			1	1
	01	m_4	m_5	m_7	m_6
	11	1	1	1	1
	10	m_{12}	m_{13}	m_{15}	m_{14}
		m_8	m_9	m_{11}	m_{10}

$$F = A'B + A'C + CD$$

		C			
		CD		11	10
A	AB	00	01	11	10
		m_0	m_1	m_3	m_2
	00	0	0		
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	0	0		0
		m_8	m_9	m_{11}	m_{10}

$$F' = B'C' + AC' + AD'$$

$$F = (B+C)(A'+C)(A'+D)$$

(c) $A'B'D' + A'D + BCD + BC'D'$

		C			
		CD		11	10
A	AB	00	01	11	10
		m_0	m_1	m_3	m_2
	00	1	1	1	1
	01	m_4	m_5	m_7	m_6
	11	1	1	1	
	10	m_{12}	m_{13}	m_{15}	m_{14}
		m_8	m_9	m_{11}	m_{10}

$$F = A'C' + A'B' + BC'D' + BCD$$

		C			
		CD		11	10
A	AB	00	01	11	10
		m_0	m_1	m_3	m_2
	00				
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	0	0	0	0
		m_8	m_9	m_{11}	m_{10}

$$F' = AB' + AC'D + BCD'$$

$$F = (A' + B)(A' + C + D')(B' + C' + D)$$

(d) $A'B'C + A'C'D + A'BC + ABD$

		C			
		CD		11	10
A	AB	00	01	11	10
		m_0	m_1	m_3	m_2
	00		1	1	1
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	1	1		
		m_8	m_9	m_{11}	m_{10}

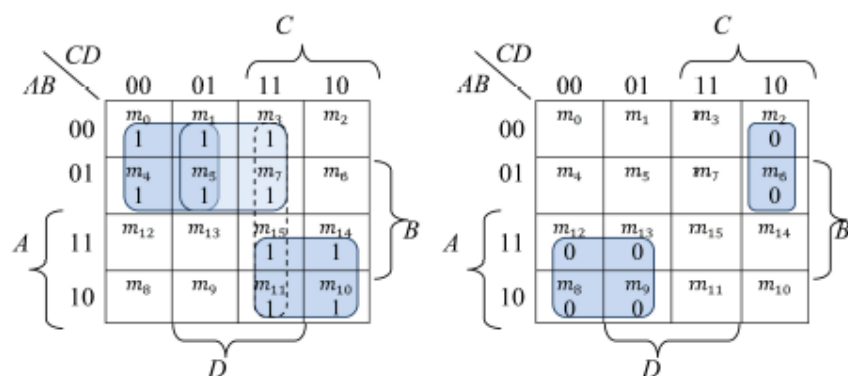
$$F = A'D + A'C + BD$$

		C			
		CD		11	10
A	AB	00	01	11	10
		m_0	m_1	m_3	m_2
	00	0			
	01	m_4	m_5	m_7	m_6
	11	m_{12}	m_{13}	m_{15}	m_{14}
	10	0	0	0	0
		m_8	m_9	m_{11}	m_{10}

$$F' = AB' + C'D' + ACD'$$

$$F = (A' + B)(C + D)(A' + C' + D)$$

3.14



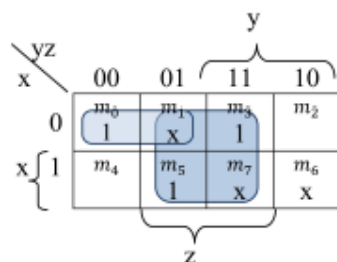
SOP form (using 1s): $F = A'C' + A'D + AC$

Alternative SOP form: $F = A'C' + AC + CD$

POS form (using 0s): $F' = AC' + A'CD'$

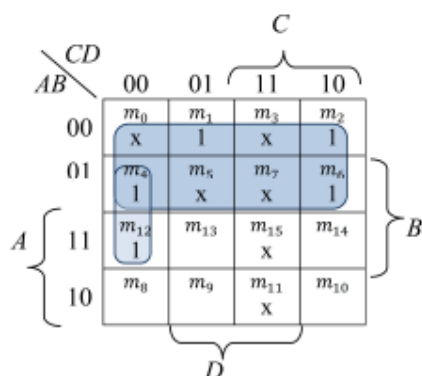
$$F = (A' + C)(A + C' + D)$$

3.15 (a) $F(x, y, z) = \Sigma(0, 3, 5)$ $d(x, y, z) = \Sigma(1, 6, 7)$



$$F(x, y, z) = xy' + z$$

(b) $F(A, B, C, D) = \Sigma(1, 2, 4, 6, 12)$ $d(A, B, C, D) = \Sigma(0, 3, 5, 7, 11, 15)$



$$F(A, B, C, D) = A' + BC'D'$$

(c) $F(A, B, C, D) = \Sigma(3, 5, 6, 11)$ $d(A, B, C, D) = \Sigma(4, 7, 9, 12, 15)$

		C			
		00	01	11	10
A	00	m_0	m_1	m_3 1	m_2
	01	m_4 x	m_5 1	m_7 x	m_6 1
	11	m_{12} x	m_{13}	m_{15} x	m_{14}
	10	m_8	m_9 x	m_{11} 1	m_{10}

$$F(A, B, C, D) = A'B + CD$$

(d) $F(A, B, C, D) = \Sigma(4, 5, 8, 9, 14, 15)$ $d(A, B, C, D) = \Sigma(2, 3, 12, 13)$

		C			
		00	01	11	10
A	00	m_0	m_1	m_3 x	m_2 x
	01	m_4 1	m_5 1	m_7	m_6
	11	m_{12} x	m_{13} x	m_{15} 1	m_{14} 1
	10	m_8 1	m_9 1	m_{11}	m_{10}

$$F(A, B, C, D) = AC' + BC' + AB$$

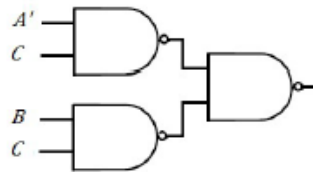
3.16 (a) $F(A, B, C, D) = A'B'C + A'BC + ABC$

		C			
		00	01	11	10
A	00	m_0	m_1	m_3 1	m_2 1
	01	m_4	m_5	m_7 1	m_6 1
	11	m_{12}	m_{13}	m_{15} 1	m_{14} 1
	10	m_8	m_9	m_{11}	m_{10}

$$F = A'C + BC$$

$$= ((A'C + BC))' = ((A'C)'(BC))'$$

$$= (A' \text{ NAND } C) \text{ NAND } (B \text{ NAND } C)$$



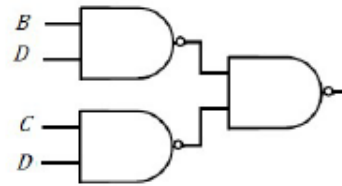
(b) $F(A,B,C,D) = A'CD + A'BD + ABD + AB'CD$

		C			
		00	01	11	10
A	00	m ₀	m ₁	m ₂ 1	m ₃
	01	m ₄	m ₅ 1	m ₇ 1	m ₆
	11	m ₁₂	m ₁₃ 1	m ₁₅ 1	m ₁₄
	10	m ₈	m ₉	m ₁₁ 1	m ₁₀

D

$$F = BD + CD$$

$$= ((BD + CD)')' = ((BD)')(CD)')$$



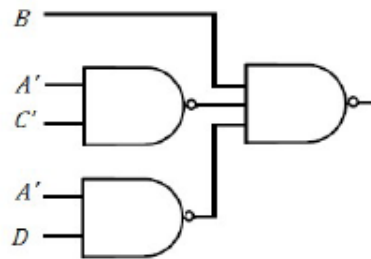
(c) $F(A,B,C,D) = (A' + B' + C)(A' + B' + C')(B' + C' + D)$

$$F = ABC' + ABC + BCD'$$

		C			
		00	01	11	10
A	00	m ₀ 1	m ₁ 1	m ₃ 1	m ₂ 1
	01	m ₄ 1	m ₅ 1	m ₇ 1	m ₆ 0
	11	m ₁₂ 0	m ₁₃ 0	m ₁₅ 0	m ₁₄ 0
	10	m ₈ 1	m ₉ 1	m ₁₁ 1	m ₁₀ 1

D

$$F = B + A'C' + A'D$$

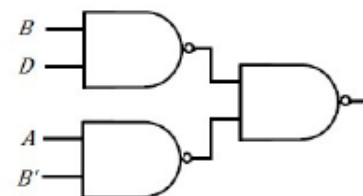


(d) $F(A,B,C,D) = AB'D' + AD + A'BD$

		C			
		00	01	11	10
A	00	m ₀	m ₁	m ₃	m ₂
	01	m ₄	m ₅ 1	m ₇ 1	m ₆
	11	m ₁₂	m ₁₃ 1	m ₁₅ 1	m ₁₄
	10	m ₈ 1	m ₉	m ₁₁	m ₁₀ 1

D

$$F = BD + AB'$$



3.17 (a) $F(A, B, C, D) = \Sigma(0, 2, 4, 5, 8, 9, 10, 11)$

$F' = \Sigma(1, 3, 6, 7, 12, 13, 14, 15)$

CD \ AB		C			
		00	01	11	10
A	00	m_0 1	m_1 1	m_2 1	m_3 1
	01	m_4 1	m_5 1	m_6 1	m_7 1
	11	m_{12} 1	m_{13} 1	m_{15} 1	m_{14} 1
	10	m_8 1	m_9 1	m_{11} 1	m_{10} 1

$F = A'B'D + BC + AB$

$= ((A'B'D + BC + AB)')' = ((A'B'D)'(BC)'(AB)')'$

(b) $F(A, B, C, D) = \Sigma(0, 2, 4, 5, 8, 9, 10, 11)$

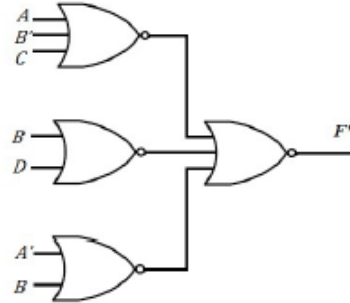
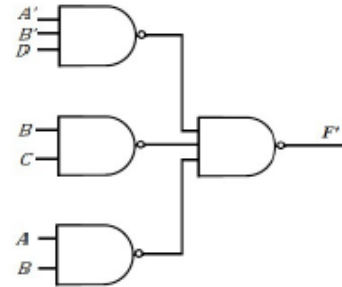
CD \ AB		C			
		00	01	11	10
A	00	m_0 1	m_1 1	m_2 1	m_3 1
	01	m_4 1	m_5 1	m_6 1	m_7 1
	11	m_{12} 1	m_{13} 1	m_{15} 1	m_{14} 1
	10	m_8 1	m_9 1	m_{11} 1	m_{10} 1

$F = B'D' + AB' + A'BC'$

$F' = (B + D)(A' + B)(A + B' + C)$

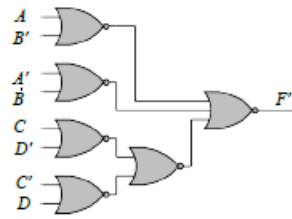
$= (((B + D)(A' + B)(A + B' + C)))'$

$= ((B + D)' + (A' + B)' + (A + B' + C)')$



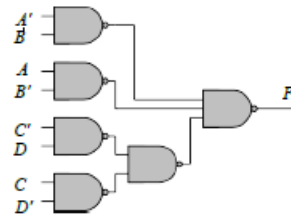
(3.18) (a)

$$F(A, B, C, D) = ((A + B')' + (A' + B)') + ((C + D')' + (C' + D)')' \quad (\text{nor gates})$$



(b)

$$F(A, B, C, D) = (A'B)' (AB')' ((C'D)' (CD'))' \quad (\text{nand gates})$$



3.19 (a) $F(w, x, y, z) = wx'y' + wy'z' + xy'$

		y			
		00	01	11	10
w	00	m_0 0	m_4 0	m_8 0	m_{12} 0
	01	m_1 1	m_5 1	m_9 0	m_{13} 0
	11	m_2 1	m_6 1	m_{10} 0	m_{14} 0
	10	m_3 1	m_7 1	m_{11} 0	m_{15} 0
		z			

$$F' = y + w'x'$$

$$F = (y + w'x')' = y'(w + x) = ((y'(w + x))')' = ((y + (w + x))')' = y \text{ NOR } (w \text{ NOR } x)$$

(b) $F(w, x, y, z) = \Sigma(0, 2, 8, 9, 10, 11, 14)$

		y			
		00	01	11	10
w	00	m_0 1	m_4 0	m_8 0	m_{12} 1
	01	m_1 0	m_5 0	m_9 0	m_{13} 0
	11	m_2 0	m_6 0	m_{10} 0	m_{14} 1
	10	m_3 1	m_7 1	m_{11} 1	m_{15} 1
		z			

$$F' = w'z + w'x + xy' + xz$$

$$F = (w + z')(w + x')(x' + y)(x' + z) = (((w + z')(w + x')(x' + y)(x' + z)))'$$

$$= ((w + z')'(w + x')'(x' + y)'(x' + z)')'$$

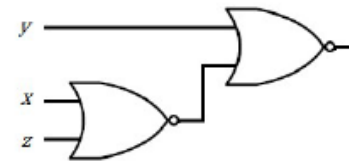
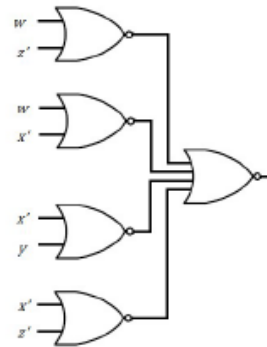
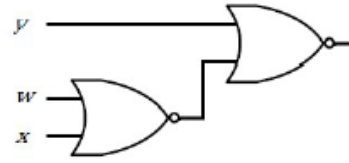
(c) $F(x, y, z) = ((x' + y)(y + z'))'$

$$F' = (x' + y)(y + z') = x'y + x'z' + y + yz'$$

		y			
		00	01	11	10
x	0	m_0 1	m_1 0	m_3 1	m_2 1
	1	m_4 0	m_5 0	m_7 1	m_6 1
		z			

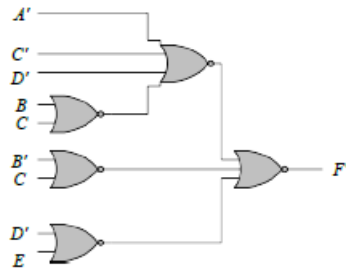
$$F' = y + x'z'$$

$$F = y'(x + z) = ((y'(x + z))')' = (y + (x + z))'$$



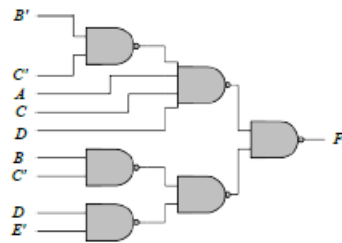
3.20 Multi-level NOR:

$$F' = [(A' + C' + D' + (B + C)')' + (B' + C)' + (D' + E)']'$$



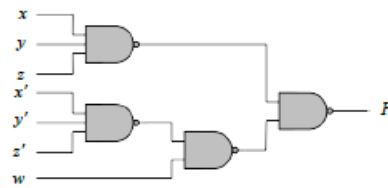
Multi-level NAND:

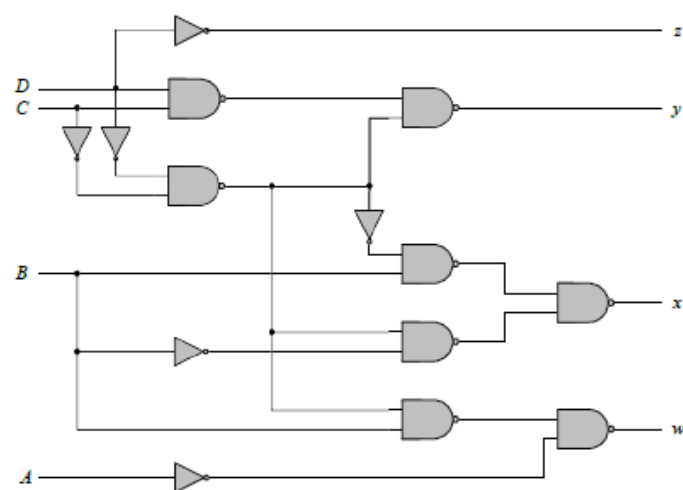
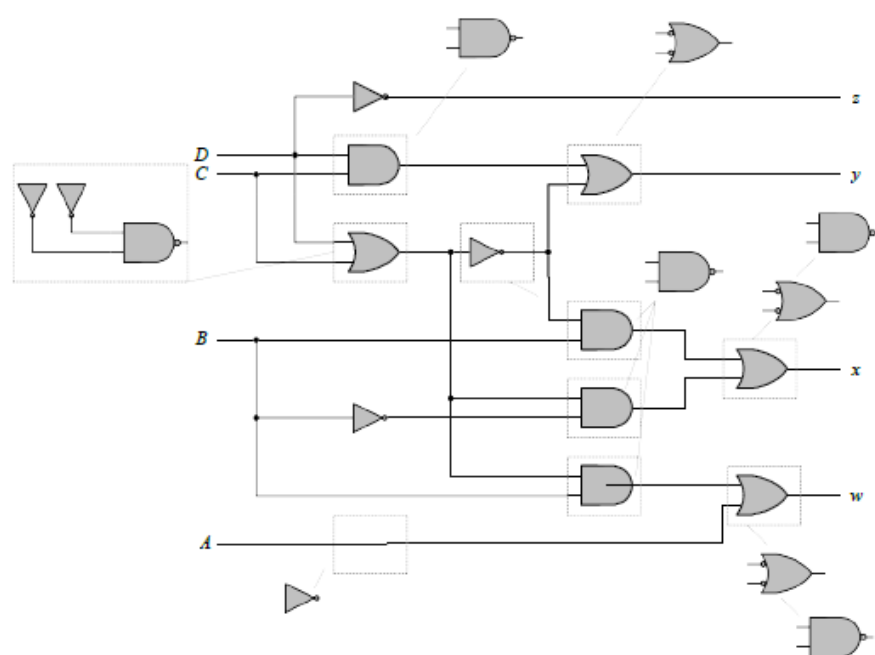
$$F' = [CD(B'C)'A]' [(BC)'(DE)']'$$



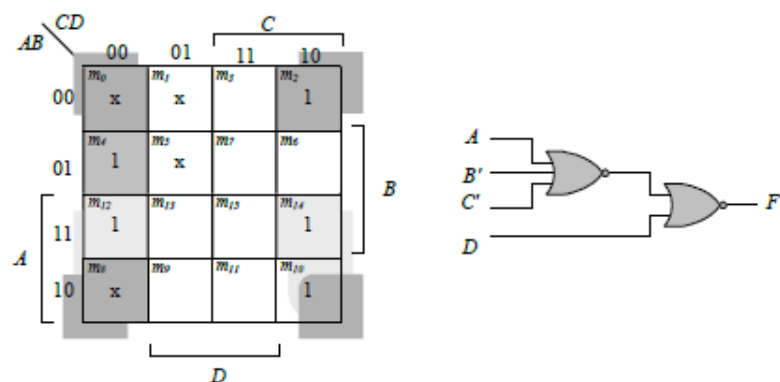
3.21

$$F' = [w(x + y + z)]' [xyz]' = [w(x'y'z')]' (xyz)'$$



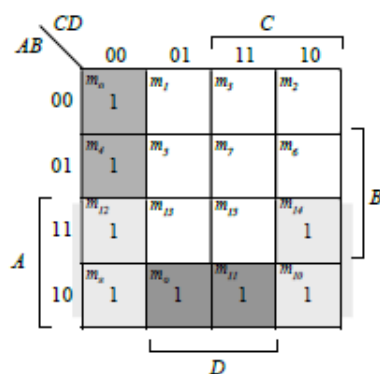


3.23

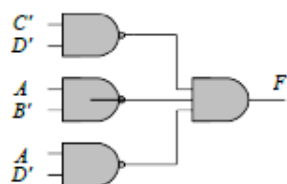


$$F = [D + A'BC]' = [D + (A + B' + C)']'$$

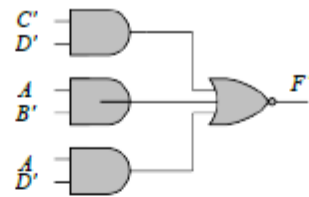
3.24 $F(A, B, C, D) = S(0, 4, 8, 9, 10, 11, 12, 14)$



(a) $F = C'D' + AB' + AD'$
 $F' = (C'D')'(AB')'(AD')'$
 AND-NAND:



(b) $F' = [C'D' + AB' + AD']'$
 AND-NOR:

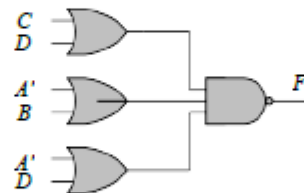


(c) $F = C'D' + AB' + AD' = (C + D)' + (A' + B)' + (A' + D)'$

$F' = (C'D')'(AB')'(AD')' = (C + D)(A' + B)(A' + D)$

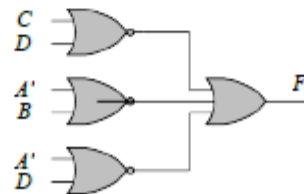
$F = [(C + D)(A' + B)(A' + D)]'$

OR-NAND:

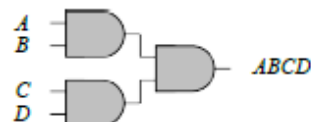


(d) $F = C'D' + AB' + AD' = (C + D)' + (A' + B)' + (A' + D)'$

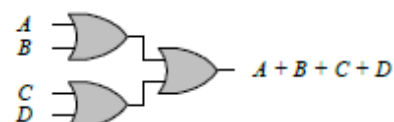
NOR-OR:



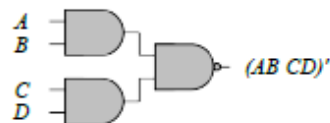
3.25



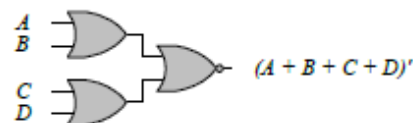
AND-AND $\rightarrow$ AND



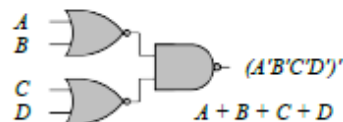
OR-OR $\rightarrow$ OR



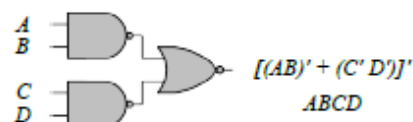
AND-NAND $\rightarrow$ NAND



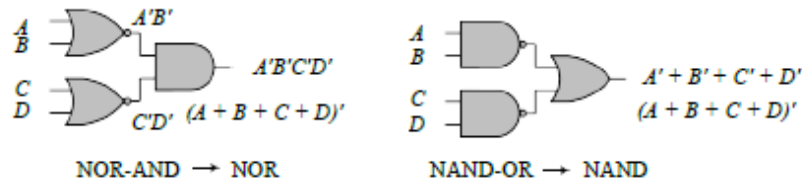
OR-NOR $\rightarrow$ NOR



NOR-NAND $\rightarrow$ OR

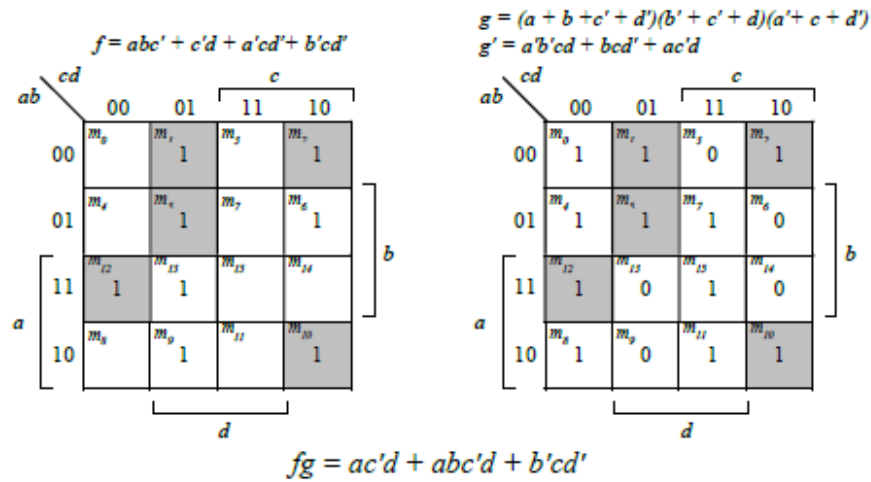


NAND-NOR $\rightarrow$ AND



The degenerate forms use 2-input gates to implement the functionality of 4-input gates.

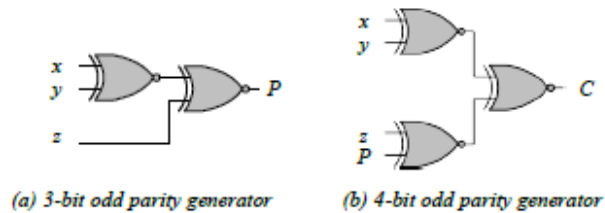
3.26



3.27

$$x \oplus y = x'y + xy'; \text{ Dual} = (x' + y)(x + y') = (x \oplus y)'$$

3.28



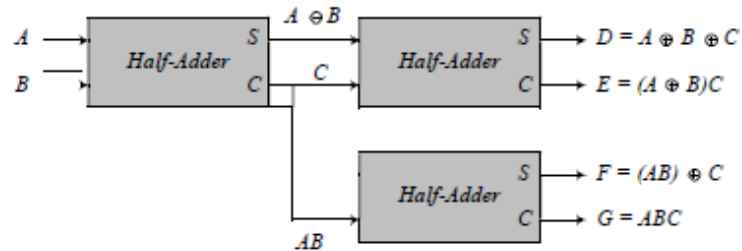
3.29

$$D = A \oplus B \oplus C$$

$$E = A'BC + AB'C = (A \oplus B)C$$

$$F = ABC' + (A' + B')C = ABC' + (AB)'C = (AB) \oplus C$$

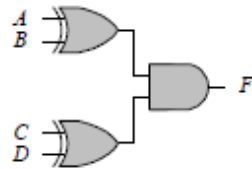
$$G = ABC$$



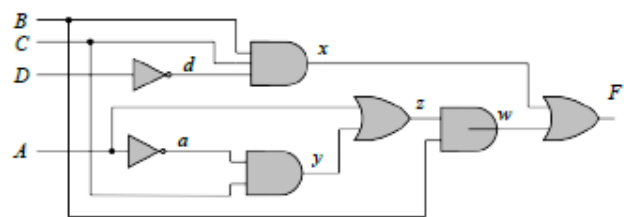
3.30

$$F = AB'CD' + A'BCD' + AB'C'D + A'BC'D$$

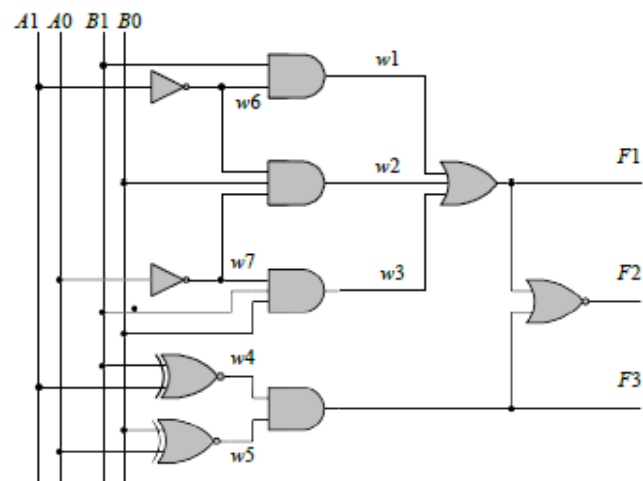
$$F = (A \oplus B)CD' + (A \oplus B)C'D = (A \oplus B)(C \oplus D)$$



3.36 (a)



(b)



(c)

